Weather- and Lighting-Robust Traffic Sign Classification
Using a Synthetic Adverse-Condition Dataset

Computer Vision & Deep Learning — Final Project Pitch

PROBLEM STATEMENT

Traffic sign recognition is a critical computer vision task underpinning driver-assistance and autonomous driving systems. Although modern deep learning models achieve high accuracy on clean benchmarks, performance degrades considerably under adverse real-world conditions — including low illumination, glare, shadows, fog, rain, and motion blur. This project addresses this reliability gap by training a robust classifier explicitly on both clean and synthetically corrupted images, enabling direct measurement of the **performance drop between clean and corrupted inputs** across multiple degradation types.

DATASET COLLECTION & ANNOTATION

Base dataset: The German Traffic Sign Recognition Benchmark (GTSRB) — 43 sign categories — serves as the clean foundation.

Synthetic adverse-condition dataset (novel contribution): Each GTSRB image is programmatically transformed via OpenCV and Albumentations to simulate seven degradation types:

1. Low brightness (night / underexposure)
2. Overexposure (intense direct sunlight)
3. Shadow overlay simulation
4. Fog — contrast reduction with white haze
5. Rain streak synthesis
6. Motion blur
7. Global contrast reduction

Each transformed image retains its original **sign-class label** and receives an additional **condition label** (clean or one of 7 corruptions), yielding a multi-label dataset suited for robustness-focused training and evaluation.

PROPOSED DEEP LEARNING SOLUTION

A pretrained ResNet-18 (ImageNet weights via torchvision) is fine-tuned as the backbone, extended with a **dual-head multi-task design**:

- **Sign Classification Head:** 43-class softmax output (primary task)
- **Condition Classification Head:** 8-class softmax — clean plus 7 corruption types (auxiliary task)

The joint loss — a weighted sum of both cross-entropy terms — forces the network to model image quality explicitly alongside sign identity. By jointly predicting both the sign class and the corruption type, the model learns **condition-aware, disentangled representations** that remain stable across unseen degradation levels, directly improving cross-condition generalization.

TRAINING & EVALUATION PLAN

Setup	Training Data	Output Heads
Baseline	Clean images only	Sign class only
Augmented	Clean + 7 corruptions	Sign class only
Multi-Task	Clean + 7 corruptions	Sign class + Condition

Evaluation metrics: Overall and per-condition classification accuracy, performance drop (clean vs. corrupted), and per-class confusion matrices on held-out test sets.

DEMO

A Google Colab-compatible Jupyter notebook will load the pre-trained model, run inference on sample images, and display each image alongside its predicted sign class, confidence score, and condition label. Results will include both successes and representative failure cases across all seven corruption types. This project demonstrates that explicit robustness training, combined with multi-task supervision, can close a critical performance gap between laboratory benchmarks and real-world deployment conditions.